

● EASY

● ADVANCED MATH

● ANSWER KEY

Advanced Math

12

10 answers · Rationale-rich · Teach from doc

Q1**C****C.** 4, 5

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear equation and the nonlinear equation shown intersect at the point 4, 5. Thus, the solution to the system is 4, 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**B****B.** $m^5q^9z^2$

Choice B is correct. Applying the commutative property of multiplication, the expression $m^4q^4z^{-1}mq^5z^3$ can be rewritten as $m^4mq^4q^5z^{-1}z^3$. For positive values of x , $x^ax^b = x^{a+b}$. Therefore, the expression $m^4mq^4q^5z^{-1}z^3$ can be rewritten as $m^{4+1}q^{4+5}z^{-1+3}$, or $m^5q^9z^2$.

Choice A is incorrect and may result from multiplying, not adding, the exponents.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**12**

The correct answer is 12. The value of $f36$ is the value of fx when $x = 36$. Substituting 36 for x in the given equation yields $f36 = 6 + \sqrt{36}$, which is equivalent to $f36 = 6 + 6$, or $f36 = 12$. Thus, the value of $f36$ is 12.

Q4**A****A. -8**

Choice A is correct. Solving for x by taking the square root of both sides of the given equation yields $x = 8$ or $x = -8$. Of the choices given, -8 satisfies the given equation.

Choice B is incorrect and may result from a calculation error when solving for x . Choice C is incorrect and may result from dividing 64 by 2 instead of taking the square root. Choice D is incorrect and may result from multiplying 64 by 2 instead of taking the square root.

Q5**D****D. $16x^2 + 9x$**

Choice D is correct. In the given expression, the first two terms, $9x^2$ and $7x^2$, are like terms. Combining these like terms yields $9x^2 + 7x^2$, or $16x^2$. It follows that the expression $9x^2 + 7x^2 + 9x$ is equivalent to $16x^2 + 9x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6**D**

D. $\frac{1}{18}$

Choice D is correct. It's given that $fx = \frac{1}{6x}$. Substituting 3 for x in this equation yields $f3 = \frac{1}{63}$, or $f3 = \frac{1}{18}$. Therefore, when $x = 3$, the value of fx is $\frac{1}{18}$.

Choice A is incorrect. This is the value of fx when $x = 0.5$.

Choice B is incorrect. This is the value of fx when $x = 1$.

Choice C is incorrect. This is the value of fx when $x = 1.5$.

Q7**15**

Accept also: -5

The correct answer is 15 or -5. By the definition of absolute value, if $x - 5 = 10$, then $x - 5 = 10$ or $x - 5 = -10$. Adding 5 to both sides of the first equation yields $x = 15$. Adding 5 to both sides of the second equation yields $x = -5$. Thus, the given equation has two possible solutions, 15 and -5. Note that 15 and -5 are examples of ways to enter a correct answer.

Q8**A**

A. $19x^2 - 133$

Choice A is correct. The expression $19x^2 - 7$ can be rewritten as $19x^2 - 197$, which is equivalent to $19x^2 - 133$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**B****B.** $5, 0$

Choice B is correct. An x -intercept of a graph in the xy -plane is a point at which the graph crosses the x -axis. The graph shown crosses the x -axis at the point $5, 0$. Therefore, the x -intercept of the graph shown is $5, 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10**B****B.** $2x - 3y > 4$

Choice B is correct. Both sides of the given inequality can be divided by 3 to yield $2x - 3y > 4$.

Choices A, C, and D are incorrect because they are not equivalent to (do not have the same solution set as) the given inequality. For example, the ordered pair $(0, -1.5)$ is a solution to the given inequality, but it is not a solution to any of the inequalities in choices A, C, or D.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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